Assignment 03

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1. Given an array of integers, count the frequency of each element using a hash table.
2. You have a hashtable where its values are integers, find the key associated with the highest value.
3. You have a hashtable , the user will enter targetValue find all keys that associated with a specific targetValue
   Note : if the targetValue not found print(“Key not found”)

   Ex:

   Input :
   { "key1", "apple" }, { "key2", "banana" }, { "key3", "apple" }
   apple
   Output:
   key1
   key3
4. Given an array of strings, group anagrams together.
5. Given an array of integers, check if the array contains any duplicates.
6. Implement a SortedDictionary that stores student IDs (int) and their names (string). Perform operations like adding, removing, and retrieving student names.
7. Create an employee directory where employee IDs (int) are keys and employee names (string) are values. Use a SortedList to manage and retrieve employees in order of their IDs.
8. Given an array of integers from 1 to N with some numbers missing, find the missing numbers.
9. You have a list of integers with possible duplicates, create a HashSet that contains only unique values.

10. You have a hashtable with unique values, create a new hashtable where the keys and values are swapped.
11. Find the union of two sets, returning the unique elements from both sets.
12. You have a dictionary with string keys, the user will enter targetChar , count how many keys start with this targetChar.

    Ex:
    Input :
    {“apple”,1},
    {“animal”,2},
    {“airport”,3}
    a
    Output:
    3

13. You have a sorted set , user will enter an integer target, find all elements that is greater than target and add them in list.
14. You have a sorted list with integer values, find all the keys associated with even values.